Adaptime: Method Overview

Thesis project

Current protocol

Abstract

Adaptime studies whether a small fitted adaptor can combine a time-series foundation model with fixed, calendar-aligned retrieval while preserving the official TIME test protocol. This note defines the current univariate `full_ridge_shared` formulation; it reports no empirical result.

1 Chronological protocol

Each dataset and forecast term is divided into a fixed datastore, adaptation training, adaptation validation, and the untouched official TIME test interval. The first three regions end strictly before the test. Ridge candidates are fit once on adaptation training, K and α are selected on adaptation validation, and the selected coefficients are frozen for test. There is no train-plus-validation refit.

Let H be the forecast horizon, T_{test} TIME's configured test length, and $n = \lfloor T_{\text{test}}/H \rfloor$ its official complete test-window count per variate. Adaptation training contains $2n$ window origins and validation contains n, giving three times the test-window count before test. Their interval lengths are $H + (2n-1)s_f$ and $H + (n-1)s_f$, where the prime stride s_f depends on frequency rather than on H: 127 for intraday/hourly, 27 for business-daily/daily, 11 for weekly, 5 for monthly, and 3 for quarterly data. The remaining earlier observations form the datastore.

The ridge history length is the same as in the inherited vanilla TIME run for its backbone: 8192 for Chronos-2, 4096 for TS-ICL, and 2048 for Chronos-Bolt. TS-RAG retains its separate native length of 512 when used as an external control. Adaptation training and validation retain only rows with this complete fixed-length history, so their retrieval distances always compare equal-width representations. The vanilla test pass instead uses every official window with all history available at that date, capped at the backbone limit.

For a query time t, let $d_{\max}$ be the fixed datastore endpoint and P the dataset period. The latest eligible neighbor date is

$$d_t = \min(t - H, d_{\max}) - \big(\min(t - H, d_{\max}) - t\big) \bmod P. \tag{1}$$

Earlier candidates walk backward from d_t by a datastore stride that is a multiple of P. This keeps the datastore fixed while aligning its phase to the query.

By default every eligible earlier datastore date is accessible. An optional global window cap is divided evenly across variates, rounded down, and retains the most recent dates. TS-RAG requires every variate to retain at least 11 datastore dates. Shared preparation records these counts; before TS-RAG representation extraction, its project-owned data adapter checks the minimum globally, including when a previously completed artifact is reused. Generic preparation remains available to the Ridge family, whose insufficient-history path is the vanilla fallback described below. An early training or validation origin that cannot supply the fixed history is excluded from fitting, but no official test origin is removed. If too few valid fixed-context training windows remain, no ridge is fitted: every method evaluates TIME's unchanged test windows with cached vanilla forecasts and records the task-level fallback reason.

Let q_t be the finite fraction of a query history. The default gate requires $q_t \geq 0.8$ before retrieval. Datastore histories satisfy the same threshold, and their complete retrieved future must be finite. A query is RAG-eligible only when it has at least $K_{\max}$ valid chronological neighbors, where $K_{\max} = 15$ for the default grid. Every candidate K uses this common query support and consumes the first K neighbors from the same ordered list; fewer than $K_{\max}$ valid neighbors makes the query ineligible for all candidates. After selection is frozen, test extraction retrieves only the selected K; a test query needs those K neighbors rather than $K_{\max}$.

2 Full shared ridge

For each query, V is the vanilla foundation forecast and C is the forecast obtained when the K retrieved neighbor trajectories are supplied as covariates. For neighbor i, Y_i is its observed horizon and N_i is the foundation model's vanilla forecast from that neighbor's own history. The full design is

$$X = [V, C, Y_1, \ldots, Y_K, N_1, \ldots, N_K]. \tag{2}$$

The paired Y_i and N_i terms expose neighbor residual information. Let $s_{t,c}$ be the root mean squared seasonal difference over the complete history before query t for channel c. Only pairs whose two original dates are finite contribute, and their count is the denominator. One no-intercept ridge vector is shared across all forecast horizons and fitted by MSSE:

$$\hat{\beta}_\alpha = \arg\min_\beta \sum_{t,c,h} \left(\frac{(Y - V)_{t,c,h} - (X\beta)_{t,c,h}}{s_{t,c}} \right)^2 + \alpha \|\beta\|_2^2, \tag{3}$$

$$V' = V + X\hat{\beta}_\alpha. \tag{4}$$

The primary setting is $K = 10$, $\alpha = 10^{-2}$. Validation searches $K \in \{1, 5, 10, 15\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. Delta, convex, per-horizon, and multivariate Adaptime variants are outside the current protocol.

Ridge fitting uses complete adaptation-training rows only; it never applies an element-wise NaN mean to the normal equations. During validation and test, an ineligible query receives the vanilla output for both C and V'. This gate uses only query and datastore information available at forecast time. Missing future labels affect metric support only and cannot select the prediction path.

3 Bayesian covariate baseline

For the selected K, each eligible adaptation-training or validation window is one paired trial. Let $z_i = 1$ when the per-window MSSE of C_i is lower than that of V_i, $z_i = 0$ when it is higher, and $z_i = 1/2$ for a tie. With a Beta(1, 1) prior, the posterior-mean probability is

$$p = \frac{1 + \sum_i z_i}{2 + n}. \tag{5}$$

This evidence is frozen before test. The soft baseline is

$$B = (1 - p)V + pC. \tag{6}$$

An ineligible test window receives V for both the hard covariate and Bayesian branches. A task-level insufficient-training fallback fixes $p = 0$ and makes all four family predictions equal to V.

4 Evaluation

The main official-test report compares vanilla V, hard retrieval-context C, Bayesian B, and frozen full Ridge V'. Each receives its own standard TIME evaluation artifact on identical official windows. The evaluator retains MSE, MAE, RMSE, MAPE, sMAPE, MASE, ND, and CRPS. Seasonal Naive remains the inherited external baseline used by broader TIME summaries. Missing dates are never removed or shifted: forecast errors skip non-finite target dates and MASE scales count only finite lagged pairs at their original timestamps. The comparison reports each metric mean together with its finite and total value counts; unequal finite coverage remains visible rather than preventing report construction. Conclusions require completed and inspected cluster artifacts.

Eligibility arrays, reason codes, effective training counts, and test RAG coverage, Ridge coefficients, and Bayesian evidence are retained with the task artifacts. The resulting adapted estimands are transparent gated hybrids: their selected method on eligible windows and vanilla otherwise. An insufficient-training task is the limiting case with zero eligible Adaptime windows and vanilla predictions for all four family branches.